Module I — Crystal structure & X-rays

Derivations, step by step

Only 1, 2, 3, 4 and 6-fold rotation axes are allowed

5 marks

A, B, C, D: lattice points on one row, T apart. An n-fold axis turns the lattice into itself by $\theta = 2\pi/n$. B', C': where A and D land after turning about B and C. m: a whole number.

BEFORE YOU START: WHAT THIS USES

$A, B, C, D,\ T$	lattice points on a row and their spacing: lattice and translation vector
n-fold axis	a turn by $2\pi/n$ that carries the lattice into itself: the symmetry operations
$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$	school trigonometry: the right triangle is drawn in the steps

START

$$\theta = \frac{2\pi}{n}$$

RESULT

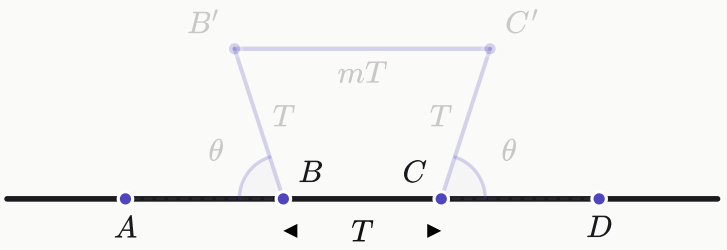
$$n = 1, 2, 3, 4, 6$$

SET-UP

$$\theta = \frac{2\pi}{n}$$

An n-fold axis means a turn by $\theta = 2\pi/n$ leaves the lattice looking the same (symmetry operations). Suppose such axes pass through the lattice points B and C, which are one step T apart (translation vector).

A row of lattice points, spacing T



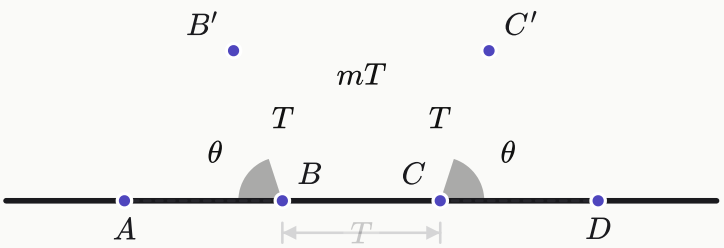
TURN ABOUT B AND C

$$B'C' \parallel BC$$

Turn about B clockwise and about C anticlockwise

Turning A about B by θ gives B'; turning D about C by θ the other way gives C'. Both are at the same height, so $B'C'$ is parallel to the row. They must be lattice points, because the turn is a symmetry.

$B'C'$ is parallel to the row



$$B'C' = mT$$

Points on a lattice row are a whole number of steps apart

B' and C' are lattice points on a row parallel to BC. Moving the whole lattice by T along that row leaves it unchanged, so the points of the row are spaced T apart, and the distance $B'C'$ is a whole number m of steps.

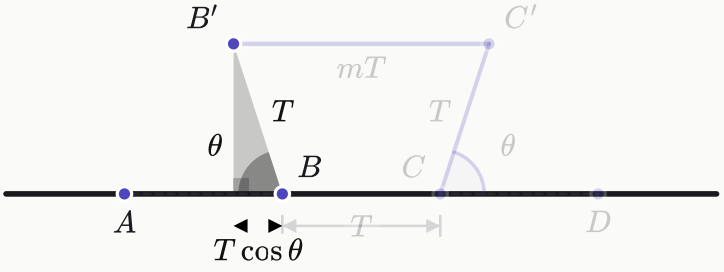
RIGHT TRIANGLE AT B

$$\cos\theta = \frac{BF}{BB'}$$

Cosine in the right triangle BFB'

Drop $B'F$ perpendicular to the row. In triangle BFB' the right angle is at F, the angle at B is θ, BF is the side next to θ and $BB' = T$ is the hypotenuse.

$\cos\theta$ = adjacent/hypotenuse = BF/BB'



Check. In the right triangle BFB', what is $\cos\theta$?

- ✓ BF/BB'
- $B'F/BB'$
- $BF/B'F$
- BB'/BF

Cosine is adjacent over hypotenuse: the adjacent side is BF, the hypotenuse is BB'.

$$BF = T\cos\theta$$

multiply: $BB' = T$

The hypotenuse BB' is T: turning does not change a length.

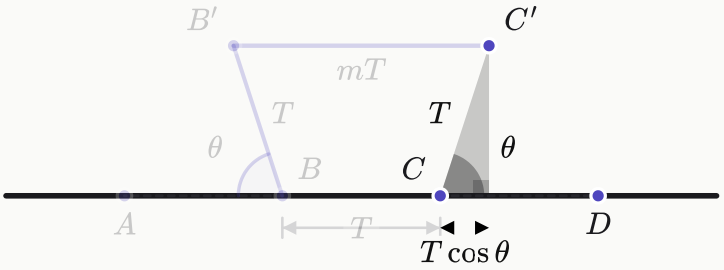
THE SAME AT C

$$CG = T\cos\theta$$

Same triangle on the other side

Triangle CGC' is the mirror image, so $CG = T\cos\theta$ too.

$CG = T\cos\theta$



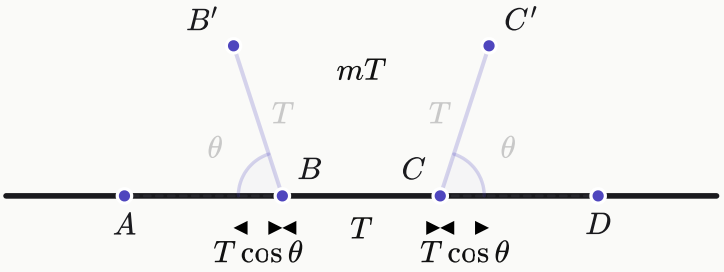
ADD THE PIECES

$$B'C' = T\cos\theta + T + T\cos\theta$$

Read $B'C'$ off the row: $FB + BC + CG$

Along the row, $B'C' = FB + BC + CG$. The $\cos\theta$ pieces lie outside BC when $\theta < 90°$; for larger θ the cosine is negative and the same formula still holds.

$B'C' = FB + BC + CG$



$$T + 2T\cos\theta = mT$$

substitute: $B'C' = mT$

Put $B'C' = mT$ on the left and tidy.

$$1 + 2\cos\theta = m$$

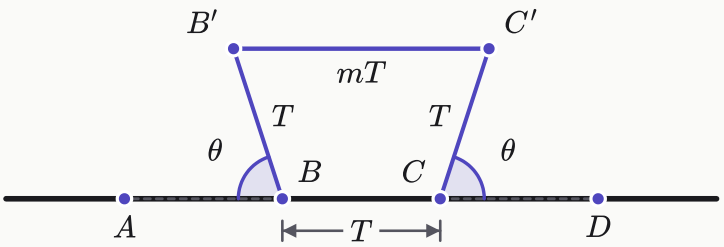
divide: T

T cancels: the result does not depend on the lattice spacing.

$$\cos\theta = \frac{m-1}{2}$$

rearrange

This is the standard form.



WHICH M ARE POSSIBLE

$$-1 \le \frac{m-1}{2} \le 1$$

A cosine lies between -1 *and* 1

So $m-1$ can only be $-2, -1, 0, 1, 2$.

Check. Which whole numbers m are allowed?

- ✓ $m = -1, 0, 1, 2, 3$
- $m = 0, 1, 2, 3, 4, 5$
- any whole number
- $m = 1, 2, 3, 4, 6$

$-1 \le (m-1)/2 \le 1$ gives $-1 \le m \le 3$.

$$\theta = 180°,\ 120°,\ 90°,\ 60°,\ 0°$$

evaluate: $m = -1, 0, 1, 2, 3$

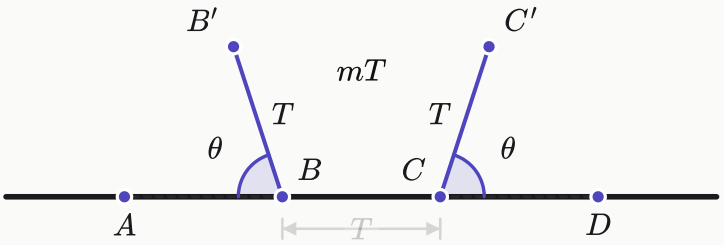
$\cos\theta = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$ for $m = -1, 0, 1, 2, 3$.

$$n = 2,\ 3,\ 4,\ 6,\ 1$$

substitute: $n = 2\pi/\theta$

No 5: a 5-fold turn is $72°$, and $\cos 72° = 0.309$ would need $m = 1.618$, not a whole number. That is why pentagons cannot tile a plane.

$\theta = 72°$ (drawn): $B'C' = 1.62\,T$, not a lattice step



Interplanar spacing d_{hkl}

5 marks

The plane (hkl) nearest the origin cuts the axes at $OA = a/h$, $OB = b/k$, $OC = c/l$. $ON = d$ is the perpendicular from the origin to the plane; α, β, γ are its angles with the axes. The axes are at right angles.

BEFORE YOU START: WHAT THIS USES

(hkl)	Miller indices, the reciprocals of the intercepts: naming a plane
$a,\ b,\ c$	lattice constants along the three axes (equal for a cube): crystal systems
d	distance between neighbouring planes of the family (hkl): interplanar spacing
$\cos\alpha,\ \cos\beta,\ \cos\gamma$	direction cosines: proved in the "Direction cosines" steps below
right triangle	trigonometry only: each triangle is built and labelled in the steps

START

$$OA = \frac{a}{h},\quad OB = \frac{b}{k},\quad OC = \frac{c}{l}$$

RESULT

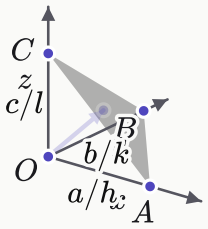
$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

INTERCEPTS

$$OA = \frac{a}{h},\quad OB = \frac{b}{k},\quad OC = \frac{c}{l}$$

Miller indices are reciprocals of the intercepts in units of a, b, c (naming a plane). So the plane next to the origin cuts the axes at $a/h,\ b/k,\ c/l$.

Intercepts $a/h,\ b/k,\ c/l$

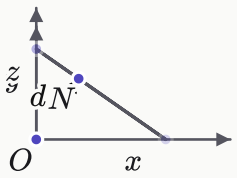


$$ON = d$$

The spacing is the perpendicular from O to the plane

The plane through O is the next one of the family, so the distance between neighbouring planes is ON.

Turned so the plane is seen edge-on: $ON \perp$ plane



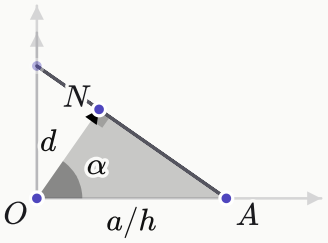
RIGHT TRIANGLE ONA

$$\cos\alpha = \frac{ON}{OA}$$

Cosine in the right triangle ONA

ON is perpendicular to the plane, so it is perpendicular to NA, which lies in the plane. The right angle is at N, the angle at O is α, and OA is the hypotenuse.

Triangle ONA flat on the screen: $\cos\alpha = ON/OA$



Check. In the right triangle ONA (right angle at N), what is $\cos\alpha$?

- ✓ ON/OA
- NA/OA
- OA/ON
- ON/NA

Adjacent over hypotenuse: next to α is ON; opposite the right angle is OA.

$$\cos\alpha = \frac{d}{a/h}$$

substitute: $ON = d$, $OA = a/h$

$$\cos\alpha = \frac{dh}{a}$$

simplify

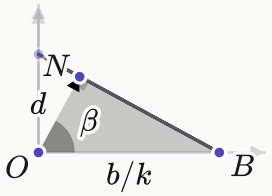
THE OTHER TWO AXES

$$\cos\beta = \frac{ON}{OB} = \frac{dk}{b}$$

Same steps in the right triangle ONB

Turn the view until triangle ONB is flat: right angle at N, angle β at O, hypotenuse $OB = b/k$.

Triangle ONB flat on the screen: $\cos\beta = ON/OB$

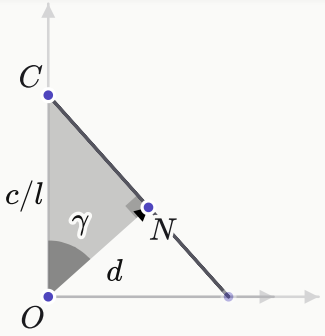


$$\cos\gamma = \frac{ON}{OC} = \frac{dl}{c}$$

And in the right triangle ONC

Hypotenuse $OC = c/l$, angle γ at O.

Triangle ONC flat on the screen: $\cos\gamma = ON/OC$



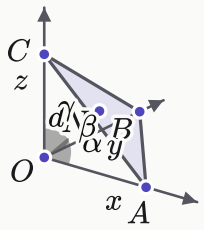
DIRECTION COSINES

$$\hat{\mathbf{n}} = (\cos\alpha,\ \cos\beta,\ \cos\gamma)$$

Components of a unit vector along the three perpendicular axes

The component of a unit vector $\hat{\mathbf{n}}$ along an axis is the cosine of the angle between them, so a unit vector along ON has the components $\cos\alpha, \cos\beta, \cos\gamma$.

Back to 3-D: one line, three angles



$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Length of the vector from its components

Pythagoras in the plane gives $n_x^2 + n_y^2$; once more with n_z gives the length squared, which is 1. This needs perpendicular axes.

$$d^2\left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}\right) = 1$$

Put in the three cosines and take out d^2

$$d^2 = \frac{1}{\dfrac{h^2}{a^2} + \dfrac{k^2}{b^2} + \dfrac{l^2}{c^2}}$$

divide: $\left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}\right)$

$$d = \frac{1}{\sqrt{\dfrac{h^2}{a^2} + \dfrac{k^2}{b^2} + \dfrac{l^2}{c^2}}}$$

root

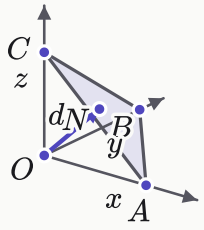
The general result for orthogonal axes (cubic, tetragonal, orthorhombic).

CUBIC CRYSTAL

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

substitute: $a = b = c$

For a cube put $a = b = c$. Learn this line: most numericals use it.



Check. For a cubic crystal, what is d_{110}?

- ✓ $a/\sqrt{2}$
- $a\sqrt{2}$
- $a/2$
- $a/\sqrt{3}$

$h^2 + k^2 + l^2 = 1 + 1 + 0 = 2$, so $d = a/\sqrt{2}$.

Packing fraction of BCC = 0.68

4 marks

Cube side a, atomic radius r. In BCC the atoms touch along the body diagonal.

BEFORE YOU START: WHAT THIS USES

n	atoms per cell: counting atoms in a cell
a, r	cube edge and atomic radius; atoms touch along the body diagonal
PF	packing fraction: the two numbers of a packing
$\frac{4}{3}\pi r^3$	sphere volume: derived in the FCC derivation
BCC	body-centred cubic lattice: Bravais lattices

START

$$n = 8 \times \tfrac{1}{8} + 1$$

RESULT

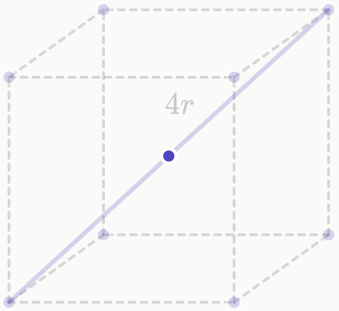
$$\text{PF} = \frac{\sqrt{3}\,\pi}{8} \approx 0.68$$

ATOMS PER CELL

$$n = 8 \times \tfrac{1}{8} + 1$$

Eight corners shared by 8 cells each, and one atom wholly inside.

$8 \times 1/8 + 1$



$$n = 2$$

evaluate: $1 + 1$

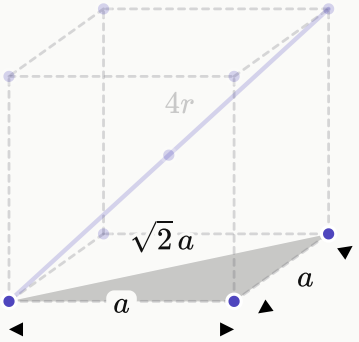
FACE DIAGONAL FIRST

$$(\text{face diagonal})^2 = a^2 + a^2$$

Pythagoras on the bottom face

The bottom face is a square of side a: the face diagonal is the hypotenuse of a right triangle with legs a, a.

Bottom face: legs a and a



$$\text{face diagonal} = \sqrt{2}\,a$$

root

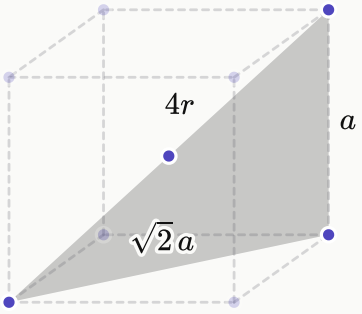
THEN THE BODY DIAGONAL

$$(4r)^2 = (\sqrt{2}\,a)^2 + a^2$$

Pythagoras in the upright triangle

The body diagonal, the face diagonal and one vertical edge form a right triangle (right angle at the corner under the top). Corner atom, body-centre atom and opposite corner atom touch along it, so it is $4r$.

$(4r)^2 = (\sqrt{2}\,a)^2 + a^2$



Check. The body diagonal is the hypotenuse of which right triangle?

- ✓ **face diagonal $\sqrt{2}a$ and edge a**
- two edges a and a
- face diagonal $\sqrt{2}a$ and $4r$
- edge a and $2r$

The vertical edge stands at right angles to the bottom face, so it is at right angles to the face diagonal.

$$4r = \sqrt{3}\,a$$

root

$2a^2 + a^2 = 3a^2$.

$$r = \frac{\sqrt{3}}{4}a$$

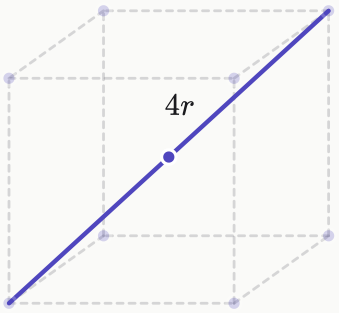
divide: 4

FILL THE CELL

$$\mathrm{PF} = \frac{2 \cdot \frac{4}{3}\pi}{a^3}\left(\frac{\sqrt{3}\,a}{4}\right)^3$$

Volume of 2 atoms ÷ a^3

2 atoms in a^3



$$\mathrm{PF} = \frac{8\pi}{3} \cdot \frac{3\sqrt{3}}{64}$$

simplify

$$\mathrm{PF} = \frac{\sqrt{3}\,\pi}{8} \approx 0.68$$

simplify

Compare: SC $\pi/6 = 0.52$ (atoms touch along an edge, $a = 2r$), BCC 0.68, FCC and HCP 0.74. Coordination numbers 6, 8, 12.

Packing fraction of FCC = 0.74

4 marks

Cube side a, atomic radius r. Packing fraction = volume filled by atoms in one cell ÷ volume of the cell.

BEFORE YOU START: WHAT THIS USES

n	atoms per cell: corners shared by 8 cells, faces by 2: counting atoms in a cell
a, r	cube edge and atomic radius (atoms taken as touching hard spheres)
PF	packing fraction, volume of atoms over volume of the cell: the two numbers of a packing
$\frac{4}{3}\pi r^3$	volume of a sphere: derived in the step "Volume of one atom" below
FCC	face-centred cubic lattice: Bravais lattices

START

$$n = 8 \times \tfrac{1}{8} + 6 \times \tfrac{1}{2}$$

RESULT

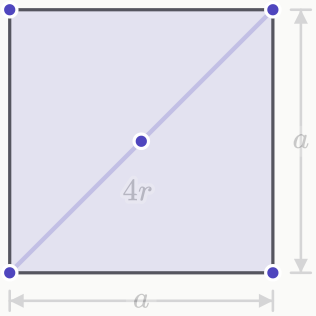
$$\mathrm{PF} = \frac{\pi}{3\sqrt{2}} \approx 0.74$$

ATOMS PER CELL

$$n = 8 \times \tfrac{1}{8} + 6 \times \tfrac{1}{2}$$

Each corner atom is shared by 8 cells, each face-centre atom by 2 cells (counting the atoms inside one cell).

Corners count $1/8$, face centres $1/2$



$$n = 4$$

evaluate: $1 + 3$

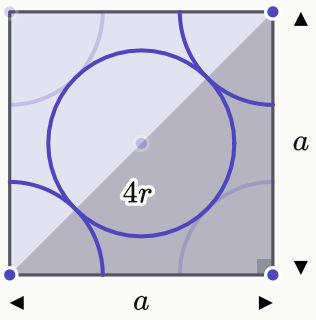
WHERE THEY TOUCH

$$(4r)^2 = a^2 + a^2$$

Pythagoras on half the face

Along the face diagonal the corner atom, the face-centre atom and the next corner atom touch: the diagonal is $r + 2r + r = 4r$. The two cube edges are the other sides of the right triangle.

$(\text{face diagonal})^2 = a^2 + a^2$



Check. Which right triangle gives the face diagonal?

- ✓ legs a and a, hypotenuse $4r$
- legs a and $4r$, hypotenuse $a\sqrt{2}$
- legs $2r$ and $2r$, hypotenuse a
- legs a and $2r$, hypotenuse $4r$

The diagonal $4r$ is opposite the right angle at the cube corner; the legs are two edges of length a.

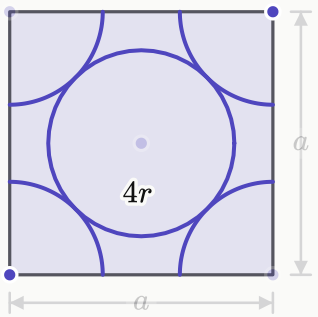
$$4r = \sqrt{2}\,a$$

root

$$r = \frac{a}{2\sqrt{2}}$$

divide: 4

$r = a/2\sqrt{2}$



VOLUME OF ONE ATOM

$$V_{\text{atom}} = \int_{-r}^{r} \pi\left(r^2 - z^2\right) dz$$

Slice the sphere into thin discs

At height z the slice is a disc of radius $\sqrt{r^2 - z^2}$ (Pythagoras), area $\pi(r^2 - z^2)$, thickness dz.

$$V_{\text{atom}} = \pi\left[r^2 z - \frac{z^3}{3}\right]_{-r}^{r} = \tfrac{4}{3}\pi r^3$$

evaluate

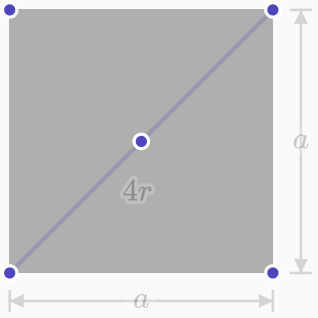
$\pi\left(2r^3 - \tfrac{2}{3}r^3\right) = \tfrac{4}{3}\pi r^3$.

FILL THE CELL

$$\text{PF} = \frac{n \cdot \frac{4}{3}\pi r^3}{a^3}$$

Volume of atoms ÷ volume of cell

atoms' volume / a^3



$$\text{PF} = \frac{4 \cdot \frac{4}{3}\pi}{a^3}\left(\frac{a}{2\sqrt{2}}\right)^3$$

substitute: $n = 4,\ r = a/2\sqrt{2}$

$$\mathrm{PF} = \frac{16\pi}{3} \cdot \frac{1}{16\sqrt{2}}$$

simplify

$(2\sqrt{2})^3 = 16\sqrt{2}$, and a^3 cancels.

$$\mathrm{PF} = \frac{\pi}{3\sqrt{2}} \approx 0.74$$

simplify

74% of the space is filled: FCC (and HCP) is the closest packing of equal spheres. Coordination number 12.

Ideal HCP: $c/a = \sqrt{8/3} = 1.633$

5 marks

P, Q, R: three touching atoms of layer A, side a. T: the layer-B atom resting in their hollow. G: centre of triangle PQR, right below T. M: midpoint of PQ. Layers A and B are $c/2$ apart.

BEFORE YOU START: WHAT THIS USES

P, Q, R, T	three touching atoms of layer A and the layer-B atom in their hollow: close packing and c/a
$a = 2r$	centre-to-centre distance of touching atoms
c	height of the HCP cell, two layers: close packing and c/a
$\sin 60°$	not assumed: its value comes out of the first steps

START

$$PM = \frac{a}{2}$$

RESULT

$$\frac{c}{a} = \sqrt{\frac{8}{3}} \approx 1.633$$

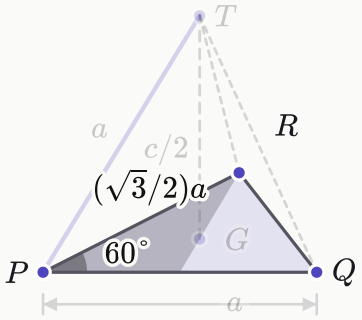
THE MEDIAN OF THE TRIANGLE

$$PM = \frac{a}{2}$$

The median RM meets PQ at its midpoint, at a right angle

PQR is equilateral, so the line from R to the midpoint M of PQ is also perpendicular to PQ. It cuts the triangle into two equal right triangles.

RM is the median of the triangle



$$RM^2 = PR^2 - PM^2$$

Pythagoras in the right triangle PMR

The right angle is at M; the hypotenuse is PR.

Check. In the right triangle PMR (right angle at M), which side is the hypotenuse?

- ✓ PR
- RM
- PM
- PQ

The hypotenuse is opposite the right angle at M: that is PR.

$$RM^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$$

substitute: $PR = a,\ PM = a/2$

$$RM = \frac{\sqrt{3}}{2}a$$

root

This is $a \sin 60°$: the value $\sin 60° = \sqrt{3}/2$ comes from this triangle.

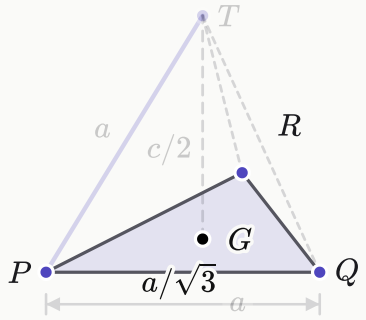
CENTRE OF THE TRIANGLE

$$PG = RG$$

T touches P, Q and R equally

The layer-B atom T sits right above a point G of layer A and touches all three atoms, so $TP = TQ = TR$. T is vertically above G, so G is the same distance from the three corners (Pythagoras in each right triangle). By symmetry it lies on the median RM.

G is equally far from P, Q, R



$$GM = RM - PG$$

G lies on the median

$RM = RG + GM$ and $RG = PG$.

$$PG^2 = PM^2 + GM^2$$

Pythagoras in the right triangle PMG

The right angle is at M.

$$PG^2 = \frac{a^2}{4} + \left(\frac{\sqrt{3}}{2}a - PG\right)^2$$

substitute: $PM = a/2,\ GM = \frac{\sqrt{3}}{2}a - PG$

$$0 = a^2 - \sqrt{3}\,a\,PG$$

Expand the bracket: PG^2 *cancels from both sides*

$\frac{a^2}{4} + \frac{3a^2}{4} = a^2$, and the cross term is $-\sqrt{3}\,a\,PG$.

$$PG = \frac{a}{\sqrt{3}}$$

rearrange

The familiar "two-thirds of the median", $\frac{2}{3} \cdot \frac{\sqrt{3}}{2}a$, now proved.

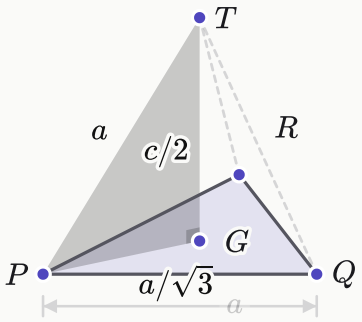
RIGHT TRIANGLE PGT

$$PT^2 = PG^2 + GT^2$$

Pythagoras in triangle PGT

T is straight above G, so GT is vertical and PG horizontal: right angle at G. PT is the hypotenuse.

$PT^2 = PG^2 + GT^2$



Check. Which side is the hypotenuse of triangle PGT?

- ✓ PT, joining the touching atoms
- GT, the height
- PG
- PQ

The right angle is at G; the side opposite it is PT.

$$a^2 = \frac{a^2}{3} + \frac{c^2}{4}$$

substitute: $PT = a,\ PG = a/\sqrt{3},\ GT = c/2$

T touches P, so $PT = 2r = a$. The layer spacing is half the cell height, $GT = c/2$.

$$\frac{c^2}{4} = \frac{2a^2}{3}$$

subtract: $\frac{a^2}{3}$

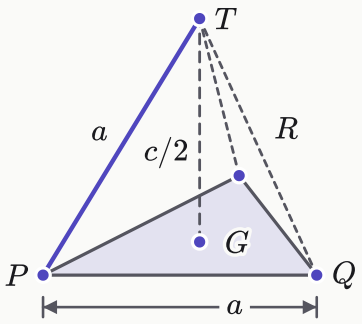
$$\frac{c^2}{a^2} = \frac{8}{3}$$

multiply: $\frac{4}{a^2}$

$$\frac{c}{a} = \sqrt{\frac{8}{3}} \approx 1.633$$

root

Real HCP metals are close: Mg 1.62, Co 1.62; Zn (1.86) and Cd (1.89) are far from ideal.



Bragg's law, $2d\sin\theta = n\lambda$

4–5 marks

Planes (hkl) are a distance d apart. θ is the glancing angle, measured from the plane. Δ is the path difference between the two reflected rays, n the order, λ the wavelength.

BEFORE YOU START: WHAT THIS USES

- d_{hkl} distance between neighbouring planes: interplanar spacing, Miller indices
- λ X-ray wavelength, comparable with atomic spacings (about 0.1 nm): diffraction methods
- θ glancing angle, measured from the plane (not the normal)
- Δ path difference of the two reflected rays; its link to phase is derived in the "Phase difference" steps

START

$$\Delta = PB + BQ$$

RESULT

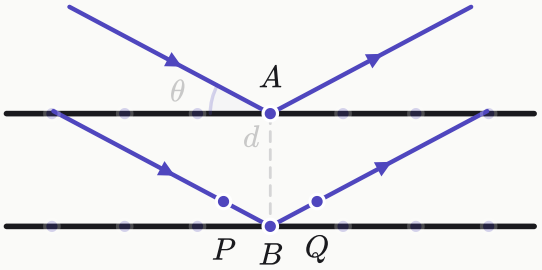
$$2d\sin\theta = n\lambda$$

EXTRA PATH

$$\Delta = PB + BQ$$

Before AP the two rays are level, and after AQ they are level again (both lines are wavefronts). So ray 2 travels the extra distance $PB + BQ$ and nothing else.

Ray 2 goes further by $PB + BQ$



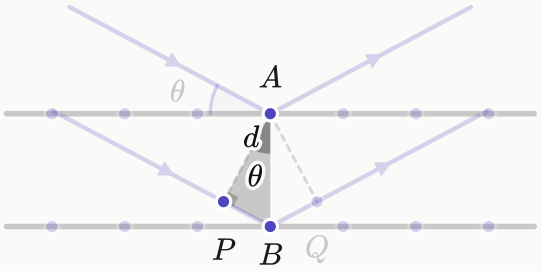
RIGHT TRIANGLE APB

$$\sin\theta = \frac{PB}{AB}$$

Sine in the right triangle APB

In triangle APB the right angle is at P. The angle at A is θ: AB is normal to the planes, AP is normal to the ray, and the angle between two lines equals the angle between their normals. PB is opposite θ and AB is the hypotenuse.

$\sin\theta$ = opposite/hypotenuse = PB/AB



Check. In the right triangle APB, what is $\sin\theta$?

- ✓ PB/AB
- AP/AB
- PB/AP
- AB/PB

Sine is opposite over hypotenuse. Opposite θ is PB; the hypotenuse is AB.

$$AB\sin\theta = PB$$

multiply: AB

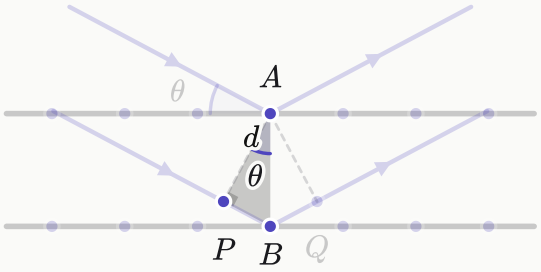
Clear the fraction.

$$PB = d\sin\theta$$

Put $AB = d$, the spacing

A and B are on neighbouring planes along the normal, so $AB = d$.

$PB = d\sin\theta$



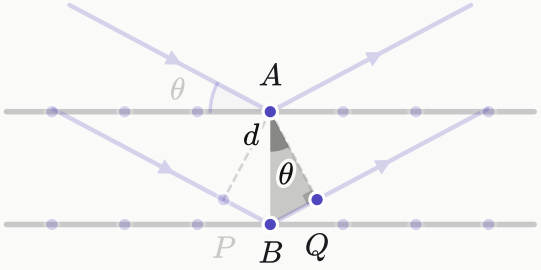
TRIANGLE AQB

$$BQ = d\sin\theta$$

Same steps in triangle AQB

Triangle AQB is the mirror image of APB: right angle at Q, angle θ at A, hypotenuse $AB = d$. So $BQ = d\sin\theta$ as well.

$BQ = d\sin\theta$, the mirror image



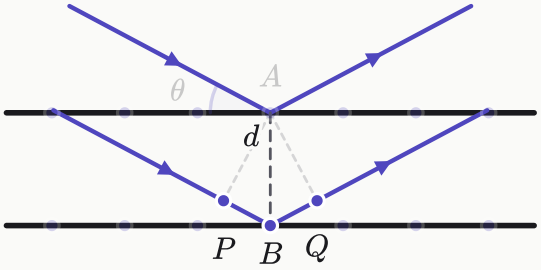
PATH DIFFERENCE

$$\Delta = d\sin\theta + d\sin\theta$$

substitute: PB, BQ

Put both pieces into $\Delta = PB + BQ$.

$\Delta = PB + BQ$



$$\Delta = 2d\sin\theta$$

combine

Two equal pieces.

PHASE DIFFERENCE

$$\varphi = \frac{2\pi}{\lambda}\,\Delta$$

A path difference Δ delays the wave by Δ/λ cycles

One wavelength of extra path is one full cycle, which is 2π of phase. So an extra path Δ is $2\pi\Delta/\lambda$ of phase.

$$\varphi = 2\pi n$$

Two waves add to a maximum when their crests coincide

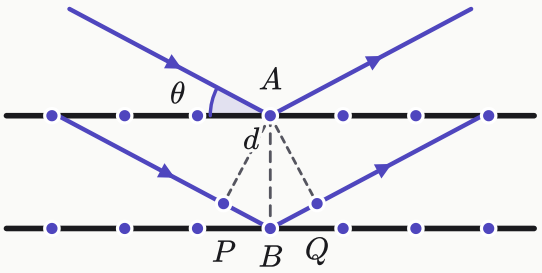
The reflected waves are added. The sum is largest when they are in step, that is when $\varphi = 0, 2\pi, 4\pi, \ldots$, a whole number n of cycles apart.

BRIGHT REFLECTION

$$2d\sin\theta = n\lambda$$

Put $\Delta = 2d\sin\theta$ into $\varphi = 2\pi n$: bright when the path difference is a whole number of wavelengths

The reflected rays reinforce only when they arrive in step, that is when Δ is a whole number of wavelengths, $n = 1, 2, 3, \ldots$ This is Bragg's law. Remember θ is from the plane, not from the normal.



Check. When do the two reflected rays reinforce each other?

- ✓ $\Delta = n\lambda$
- $\Delta = (n + \frac{1}{2})\lambda$
- $\Delta = \lambda/2$
- $\Delta = d$

Constructive interference needs the waves in phase: a path difference of a whole number of wavelengths.

Laue's equations, and Bragg's law from them

5–8 marks

A, B: identical scattering atoms, $\mathbf{r} = \overrightarrow{AB}$. $\hat{\mathbf{n}}_1$, $\hat{\mathbf{n}}_2$: unit vectors along the incident and scattered beams; the angle between them is 2θ. $\mathbf{N} = \hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$. α, β, γ: angles of $\mathbf{N}$ with the axes $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

BEFORE YOU START: WHAT THIS USES

$\mathbf{a},\ \mathbf{b},\ \mathbf{c}$	primitive vectors of the lattice: translation vectors
$h,\ k,\ l$	whole numbers, the Miller indices of the reflecting planes: naming a plane
$\cos\alpha : \cos\beta : \cos\gamma$	direction cosines of the plane normal: proved in the d_{hkl} derivation
$\frac{2\pi}{\lambda}\Delta$	phase difference: derived in Bragg's law
$\mathbf{r}\cdot\hat{\mathbf{n}}$	length of the shadow of $\mathbf{r}$ on the beam direction: shown in step 2

START

$$\Delta = AM - AL$$

RESULT

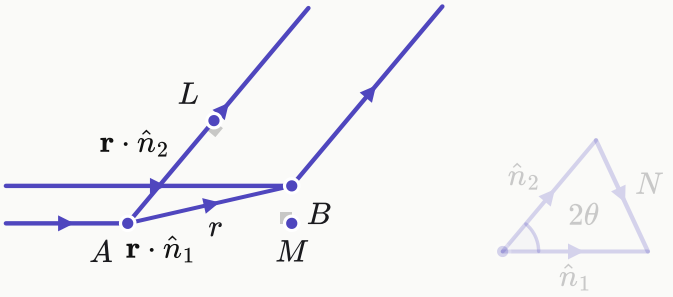
$$2d\sin\theta = n\lambda$$

PATH DIFFERENCE

$$\Delta = AM - AL$$

Projecting B onto the two beams through A: the ray through B is ahead by AM on the way in and behind by AL on the way out.

$\Delta = AM - AL$



$$\Delta = \mathbf{r}\cdot\hat{\mathbf{n}}_1 - \mathbf{r}\cdot\hat{\mathbf{n}}_2$$

A projection is a dot product with a unit vector

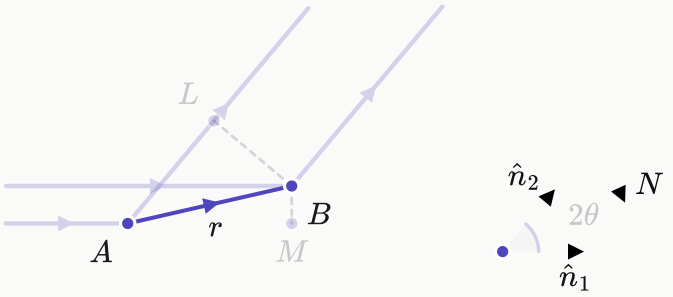
If φ is the angle between $\mathbf{r}$ and the unit vector $\hat{\mathbf{n}}$, the shadow of $\mathbf{r}$ on that line has length $|\mathbf{r}|\cos\varphi$. The dot product is $\mathbf{r}\cdot\hat{\mathbf{n}} = |\mathbf{r}|\,|\hat{\mathbf{n}}|\cos\varphi = |\mathbf{r}|\cos\varphi$, since $|\hat{\mathbf{n}}| = 1$.

$$\Delta = \mathbf{r}\cdot\mathbf{N}$$

factor

with $\mathbf{N} = \hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$.

$\mathbf{N} = \hat{n}_1 - \hat{n}_2$



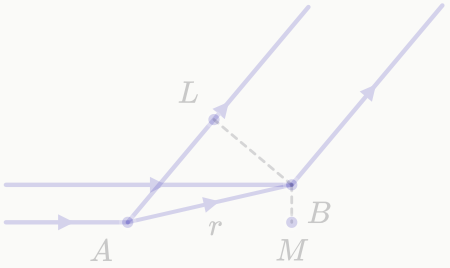
SIZE OF N

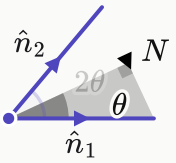
$$\sin\theta = \frac{|\mathbf{N}|/2}{1}$$

Sine in half of the isosceles triangle

$\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$ both have length 1 and meet at 2θ. The bisector cuts the triangle into two right triangles, each with hypotenuse 1, angle θ and opposite side $|\mathbf{N}|/2$.

$\sin\theta = (|\mathbf{N}|/2)/1$





Check. In the half triangle (hypotenuse 1, angle θ), what is the side opposite θ?

- ✓ $|\mathbf{N}|/2$
- $|\mathbf{N}|$
- $\cos\theta$
- $1/2$

The bisector meets $\mathbf{N}$ at its midpoint, so the opposite side is half of $|\mathbf{N}|$.

$$|\mathbf{N}| = 2\sin\theta$$

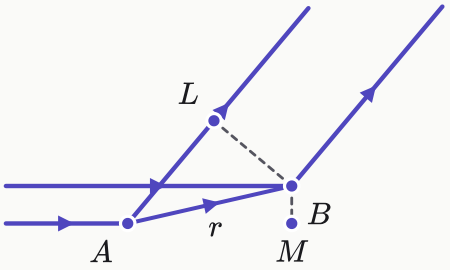
multiply: $\mathbf{2}$

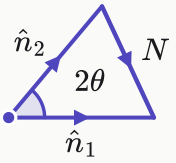
PHASE CONDITION ALONG EACH AXIS

$$\frac{2\pi}{\lambda}\,\mathbf{a}\cdot\mathbf{N} = 2\pi n h$$

Bright only if the phase difference is a whole multiple of 2π

Phase difference $= \frac{2\pi}{\lambda}\Delta$ (Bragg's law derivation). Taking $\mathbf{r}$ along the axis $\mathbf{a}$ to the next atom, it must be 2π times a whole number, written nh (the integers h, k, l have no common factor; n is that factor).





$$\mathbf{a}\cdot\mathbf{N} = nh\lambda$$

multiply: $\frac{\lambda}{2\pi}$

and in the same way $\mathbf{b}\cdot\mathbf{N} = nk\lambda$, $\mathbf{c}\cdot\mathbf{N} = nl\lambda$: all three at once.

$$2a\sin\theta\cos\alpha = nh\lambda$$

substitute: $\mathbf{a}\cdot\mathbf{N} = a\,|\mathbf{N}|\cos\alpha$

These three, with $b\cos\beta$ and $c\cos\gamma$, are Laue's equations (2.4). With $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ they fix the directions of the diffracted beams.

BACK TO BRAGG

$$\cos\alpha = \frac{nh\lambda}{2a\sin\theta}$$

divide: $2a\sin\theta$

For fixed θ, $\cos\alpha : \cos\beta : \cos\gamma = h/a : k/b : l/c$: the direction cosines of the normal to the plane (hkl). So $\mathbf{N}$ is normal to (hkl): that plane is the reflecting plane.

$$\frac{dh}{a} = \frac{nh\lambda}{2a\sin\theta}$$

substitute: $\cos\alpha = dh/a$

From the d_{hkl} derivation, the normal to (hkl) has $\cos\alpha = dh/a$.

$$2d\sin\theta = n\lambda$$

simplify

h/a cancels: Bragg's law. Laue's atom-by-atom picture and Bragg's mirror picture give the same answer.

The reciprocal of a BCC lattice is FCC

5 marks

a: cube edge. $\hat{\imath},\hat{\jmath},\hat{k}$: unit vectors along the cube edges. $\mathbf{a}',\mathbf{b}',\mathbf{c}'$: primitive vectors of BCC. Reciprocal vectors: $\mathbf{a}^* = 2\pi\,\dfrac{\mathbf{b}'\times\mathbf{c}'}{\mathbf{a}'\cdot(\mathbf{b}'\times\mathbf{c}')}$ and cyclically.

BEFORE YOU START: WHAT THIS USES

$\mathbf{a}^*,\ \mathbf{b}^*,\ \mathbf{c}^*$	reciprocal primitive vectors, defined by the formula in the legend: the reciprocal lattice
$\mathbf{a}',\ \mathbf{b}',\ \mathbf{c}'$	primitive vectors of BCC: Bravais lattices
$V = \mathbf{a}'\cdot(\mathbf{b}'\times\mathbf{c}')$	volume of the primitive cell: base area times height (note in the volume step)
$\times$	cross product, expanded as a determinant in the steps
FCC	primitive vectors $\frac{A}{2}(\hat{\imath}+\hat{\jmath}),\ldots$: Bravais lattices

START

$$\mathbf{a}' = \tfrac{a}{2}(\hat{\imath}+\hat{\jmath}-\hat{k})$$

RESULT

$$\mathbf{a}^* = \frac{2\pi}{a}(\hat{\imath}+\hat{\jmath})$$

PRIMITIVE VECTORS OF BCC

$$\mathbf{a}' = \tfrac{a}{2}(\hat{\imath}+\hat{\jmath}-\hat{k})$$

From a corner to three of the neighbouring body centres.

$$\mathbf{b}' = \tfrac{a}{2}(-\hat{\imath}+\hat{\jmath}+\hat{k}),\quad \mathbf{c}' = \tfrac{a}{2}(\hat{\imath}-\hat{\jmath}+\hat{k})$$

define

CROSS PRODUCT

$$\mathbf{b}' \times \mathbf{c}' = \tfrac{a^2}{4}(-\hat{i} + \hat{j} + \hat{k}) \times (\hat{i} - \hat{j} + \hat{k})$$

substitute

$$\mathbf{b}' \times \mathbf{c}' = \tfrac{a^2}{4}(2\hat{i} + 2\hat{j})$$

Expand the determinant

$\hat{i}$: $1 \cdot 1 - 1 \cdot (-1) = 2$; $\hat{j}$: $1 \cdot 1 - (-1) \cdot 1 = 2$; $\hat{k}$: $(-1)(-1) - 1 \cdot 1 = 0$.

$$\mathbf{b}' \times \mathbf{c}' = \tfrac{a^2}{2}(\hat{i} + \hat{j})$$

simplify

VOLUME OF THE PRIMITIVE CELL

$$V = \mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}') = \tfrac{a}{2} \cdot \tfrac{a^2}{2}(1 + 1)$$

substitute

$|\mathbf{b}' \times \mathbf{c}'|$ is the area of the parallelogram on $\mathbf{b}'$ and $\mathbf{c}'$ and $\mathbf{a}'$ dotted with the unit normal is the height, so $\mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}')$ is the volume of the parallelepiped, which is one primitive cell.

Check. What is the volume of the BCC primitive cell?

- ✓ $a^3/2$
- a^3
- $a^3/4$
- $2a^3$

The cube holds 2 lattice points, so each primitive cell is half the cube.

$$V = \frac{a^3}{2}$$

simplify

RECIPROCAL VECTORS

$$\mathbf{a}^* = 2\pi \frac{(a^2/2)(\hat{i} + \hat{j})}{a^3/2}$$

Put both into the definition of $\mathbf{a}^*$

$$\mathbf{a}^* = \frac{2\pi}{a}(\hat{i} + \hat{j})$$

simplify

In the same way $\mathbf{b}^* = \dfrac{2\pi}{a}(\hat{j} + \hat{k})$ and $\mathbf{c}^* = \dfrac{2\pi}{a}(\hat{k} + \hat{i})$.

$$\text{BCC (cube } a) \longrightarrow \text{FCC (cube } 4\pi/a)$$

Compare with the FCC primitive vectors $\frac{A}{2}(\hat{i} + \hat{j})$, *...*

These are exactly the primitive vectors of an FCC lattice with cube edge $A = 4\pi/a$. The same working the other way gives FCC → BCC, and SC → SC with edge $2\pi/a$.

Bragg's law in the reciprocal lattice

5 marks

$\mathbf{k}, \mathbf{k}'$: incident and diffracted wave vectors, $|\mathbf{k}| = |\mathbf{k}'| = 2\pi/\lambda$. $\mathbf{G}$: a reciprocal lattice vector, normal to the planes (hkl), $|\mathbf{G}| = 2\pi n/d$. θ: glancing angle.

BEFORE YOU START: WHAT THIS USES

$\mathbf{k},\ \mathbf{k}'$	wave vectors of the incident and the diffracted beam; $\lvert\mathbf{k}\rvert = 2\pi/\lambda$ (a wave gains 2π of phase per wavelength)
$\mathbf{G}$	reciprocal lattice vector, normal to the planes (hkl): the reciprocal lattice
$\lvert\mathbf{G}\rvert = 2\pi n/d$	length of $\mathbf{G}$: the reciprocal lattice
$\mathbf{k}' = \mathbf{k} + \mathbf{G}$	derived from Laue's equations in the first steps below: Laue's equations

START

$$\mathbf{a}\cdot(\mathbf{k}'-\mathbf{k}) = -2\pi n h$$

RESULT

$$2d\sin\theta = n\lambda$$

WHERE THE CONDITION COMES FROM

$$\mathbf{a}\cdot(\mathbf{k}'-\mathbf{k}) = -2\pi n h$$

substitute: $\mathbf{k} = \frac{2\pi}{\lambda}\hat{\mathbf{n}}_1,\ \mathbf{k}' = \frac{2\pi}{\lambda}\hat{\mathbf{n}}_2$

Take Laue's first equation $\mathbf{a}\cdot\mathbf{N} = nh\lambda$ with $\mathbf{N} = \hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$ (Laue's equations). A wave of wavelength λ has $|\mathbf{k}| = 2\pi/\lambda$, so $\mathbf{k} = \frac{2\pi}{\lambda}\hat{\mathbf{n}}_1$ and $\mathbf{k}' = \frac{2\pi}{\lambda}\hat{\mathbf{n}}_2$. Then $\mathbf{a}\cdot(\mathbf{k}'-\mathbf{k}) = -\frac{2\pi}{\lambda}\mathbf{a}\cdot\mathbf{N} = -2\pi n h$.

$$(\mathbf{k}'-\mathbf{k})\cdot\mathbf{T} = 2\pi\times\text{whole number}$$

The same for $\mathbf{b}$ *and* $\mathbf{c}$*; then for every lattice vector* $\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}$

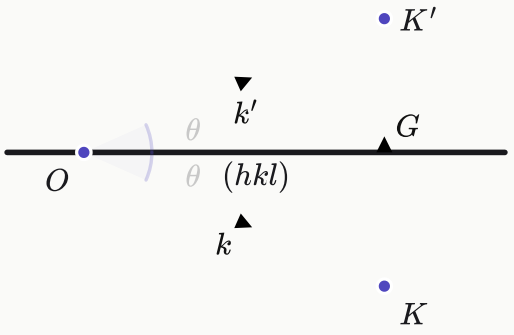
Add the three conditions with the whole-number weights n_1, n_2, n_3.

$$\mathbf{k}' = \mathbf{k} + \mathbf{G}$$

A vector whose dot product with every $\mathbf{T}$ *is* $2\pi\times$*whole number is, by definition, a reciprocal lattice vector*

Those are exactly the vectors $\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*$ of the reciprocal lattice. So a crystal can change the wave vector of an X-ray only by a reciprocal lattice vector (the sign does not matter: $-\mathbf{G}$ is one too).

$\mathbf{k}' = \mathbf{k} + \mathbf{G}$



$$k'^2 = k^2$$

Elastic scattering: the wavelength does not change

$$(\mathbf{k}+\mathbf{G})^2 = k^2$$

substitute: $\mathbf{k}' = \mathbf{k} + \mathbf{G}$

$$k^2 + 2\mathbf{k}\cdot\mathbf{G} + G^2 = k^2$$

expand

$$2\mathbf{k}\cdot\mathbf{G}+G^2=0$$

subtract: k^2

This is the diffraction condition in reciprocal space: the tip of $\mathbf{k}$ sits on the plane that bisects $\mathbf{G}$ at right angles (a Brillouin zone boundary).

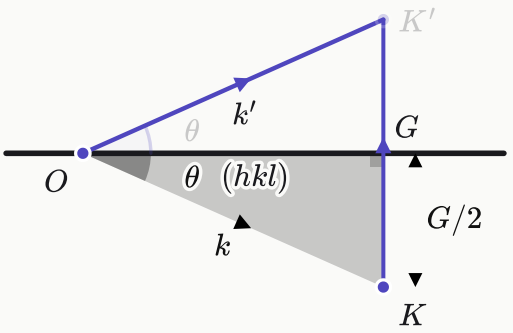
READ THE TRIANGLE

$$\sin\theta=\frac{G/2}{k}$$

Sine in half of the isosceles triangle OKK'

$\mathbf{k}$ and $\mathbf{k}'$ have equal length, so OKK' is isosceles and the plane trace OH cuts $\mathbf{G}$ in half at right angles. In triangle OHK: right angle at H, angle θ at O, opposite side $G/2$, hypotenuse k.

$\sin\theta=(G/2)/k$



Check. In the right triangle OHK, what is $\sin\theta$?

- ✓ $(G/2)/k$
- G/k
- $k/(G/2)$
- OH/k

Opposite θ is $KH=G/2$; the hypotenuse is $OK=k$.

$$G=2k\sin\theta$$

multiply: $2k$

$$\frac{2\pi n}{d}=2\cdot\frac{2\pi}{\lambda}\sin\theta$$

substitute: $G=2\pi n/d,\ k=2\pi/\lambda$

The shortest $\mathbf{G}$ normal to (hkl) has length $2\pi/d$ (reciprocal lattice); n times it gives the nth order.

$$2d\sin\theta = n\lambda$$

rearrange

Bragg's law again: the reciprocal lattice is a compact way of writing it.

